

EE-548 Silicon Photonics

Laboratory 5: Thermo-Optic Phase Modulation in a Silicon MZI

Western New England University

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Abstract

Silicon photonic Mach–Zehnder interferometers (MZIs) were investigated as optical modulators using the thermo-optic effect. In this lab, a micro-heater was integrated on one arm of an MZI to induce a phase shift via Joule heating. The goals were to characterize the phase modulation versus applied electrical power and to measure the modulator’s transfer function and key performance metrics. A broadband optical source and spectrum analyzer were used to observe interference fringes, and a source-measure unit drove the heater at incremental voltages. The induced phase shift was determined from spectral peak shifts, and the MZI output power was recorded as a function of heater bias. Results show a linear relationship between applied heater power and induced optical phase shift, achieving a π -phase shift at roughly 3 V (≈ 47 mW). The MZI’s intensity transmission followed the expected cosine-squared dependence on phase difference, with the device biased at half-wave (quadrature) for maximum linearity. A high extinction ratio of about 33 dB was obtained between the on/off states, exceeding typical values around 20 dB. These findings confirm that thermal phase modulators can provide large optical contrast, albeit with significant power consumption. Measurement limitations (e.g., fiber coupling loss and heater response time) are discussed, and improvements such as reducing thermal mass or using alternative modulation mechanisms are proposed for future work.

1 Introduction

Mach–Zehnder interferometers (MZIs) are fundamental building blocks in silicon photonics, often employed as optical modulators and switches. An MZI consists of two arms formed by splitting light from an input waveguide, then recombining the two paths at an output coupler. By inducing a phase difference between the arms (e.g., via an index change in one arm), the output intensity can be controlled through interference. In semiconductor modulators, the voltage required to switch between maximum (constructive) and minimum (destructive) interference outputs is defined as the switching voltage (commonly denoted V_π). This corresponds to a π phase difference between the light in the two arms [1]. Silicon photonics typically uses free-carrier plasma dispersion for high-speed phase modulation [1]. However, in this lab thermo-optic phase modulation is explored, which leverages silicon’s strong thermo-optic effect (large dn/dT) to change the refractive index via heating. While slower, thermo-optic MZIs are widely used for tuning and switching due to their simplicity [1].

Silicon’s refractive index increases with temperature (thermo-optic coefficient $\approx 1.86 \times 10^{-4}$ 1/K near 1550 nm [2]). Thus, a small temperature rise in one MZI arm produces a measurable phase delay relative to the unheated arm. The thermal phase shift mechanism can be described by

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta n L, \quad (1)$$

where L is the heated length and $\Delta n = (dn/dT) \Delta T$ is the index change from a temperature increase ΔT . In practice, Joule heating of a micro-resistive heater deposited on the waveguide

raises the local temperature, inducing a controlled phase delay. This phase modulation is continuous and approximately linear with applied heater power for small temperature excursions, as the index change dn/dT is roughly constant over a moderate temperature range in silicon.

A key parameter of MZI modulators is the π -phase shift—the phase change required to go from a fully destructive interference (off state) to a fully constructive interference (on state). The electrical π -power or V_π is the drive level needed to achieve this phase shift. For an ideal, lossless MZI, a π phase difference yields maximum optical transmission in one output port (and minimum in the other). Biasing the modulator at half this phase difference (i.e., at quadrature, $V_\pi/2$) is standard to obtain a linear transfer function for small-signal operation. At this quadrature bias point, the optical output versus drive voltage curve has its steepest slope, enabling the modulator to convert an electrical signal to an optical intensity modulation with minimal distortion. This is analogous to biasing an electronic amplifier at mid-point for maximum linearity.

The modulator transfer function of an MZI (neglecting loss) follows a cosine-squared dependence on the differential phase:

$$P_{\text{out}} = P_{\text{in}} \cos^2\left(\frac{\Delta\phi}{2}\right), \quad (2)$$

or equivalently $P_{\text{out}} \propto (1 + \cos \Delta\phi)/2$. A full 2π phase change brings the interference back to its initial state, producing a periodic response in drive voltage. In this experiment, starting from destructive interference (phase π at 0 V bias), a π increase in phase (to 2π total) yields constructive interference at the output. Exceeding this will begin a new interference cycle.

The extinction ratio (ER) is an important performance metric defined (in power terms) as the ratio of transmitted power in the on versus off states [3]. It is usually expressed in dB:

$$\text{ER}_{\text{dB}} = 10 \log_{10}\left(\frac{P_{\text{max}}}{P_{\text{min}}}\right). \quad (3)$$

A high ER (e.g., 20 dB or more) indicates that the modulator can nearly extinguish the light in the off state, which is crucial for digital modulation contrast [3]. Silicon MZIs typically achieve ER around 20 dB [3], though optimized designs can exceed this (e.g., > 50 dB using cascaded or balanced techniques [4]). A related metric is insertion loss, but in this experiment the focus is on relative modulation depth rather than absolute loss.

2 Theory

2.1 Thermo-Optic Phase Shift

In silicon, the dominant mechanism for thermal phase modulation is the thermo-optic effect, wherein the refractive index n increases with temperature. The material's thermo-optic coefficient is approximately $1.8 \times 10^{-4} \text{ 1/K}$ [2], meaning that even a few degrees of heating can significantly change n at telecommunication wavelengths. By locally heating one MZI arm, an index difference Δn is induced between the two arms. Over the physical path length L of the heater, this creates an optical phase delay

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta n L. \quad (4)$$

For example, if $L = 1 \text{ cm}$ and $\Delta T = 10 \text{ K}$, then

$$\Delta n \approx (1.8 \times 10^{-4}) \times 10 = 1.8 \times 10^{-3}, \quad (5)$$

and at $\lambda = 1550 \text{ nm}$,

$$\Delta\phi \approx \frac{2\pi}{1.55 \times 10^{-6}} (1.8 \times 10^{-3}) (1 \times 10^{-2}) \approx 73 \text{ rad}, \quad (6)$$

which far exceeds 2π . In practice, the devices used here employ much shorter heaters (on the order of a few millimeters or less) and smaller temperature rises, requiring on the order of 10–100 mW to achieve a π shift, as shown later. Thermal phase shifting is a slow, analog mechanism: the heater’s temperature (and thus phase) responds with a thermal time constant often on the order of tens or hundreds of microseconds [5, 6], limiting modulation speed but offering continuous tunability.

2.2 π -Phase Shift and V_π

The “ π phase shift” refers to the change in phase difference that produces the extreme swing from one interference state to the opposite. It is the basis for defining the modulator’s switching voltage V_π (or corresponding power P_π). In a thermo-optic modulator, V_π is the DC heater voltage required to induce an additional π radians of phase in one arm. Equivalently, P_π is the electrical power in the heater that yields a π shift. This value depends on the heater design, waveguide geometry, and thermal environment. Typical silicon thermo-optic phase shifters of a few millimeters length require on the order of tens to hundreds of milliwatts for a π phase change [7]. For instance, simulation of a 3 mm silicon waveguide phase shifter showed about 0.3 W needed for $\Delta\phi = \pi$ [7].

In this experiment, V_π (and P_π) is determined experimentally by finding the bias at which the MZI output that was initially at minimum rises to its maximum. Half of this drive (i.e., $V_\pi/2$) is the recommended bias point for linear modulation, known as the quadrature point, where the intensity response to small voltage variations is approximately linear.

2.3 Modulator Transfer Function

An ideal MZI’s normalized transfer function can be written as

$$T(\phi) = \frac{1 + \cos \phi}{2}, \quad (7)$$

where ϕ is the phase difference between arms. This arises from the interference of two equal-amplitude beams. If one arm is phase-modulated (e.g., $\phi = \phi_0 + \Delta\phi_{\text{heater}}$), the output intensity follows a cosine interference fringe pattern.

In this experiment, the MZI was initially set to destructive interference ($\phi_0 \approx \pi$ at the chosen laser wavelength), so zero heater voltage corresponds to a transmission minimum. As heater power increases, $\Delta\phi_{\text{heater}}$ increases roughly linearly, and the output power rises following

$$P_{\text{out}} \propto \frac{1 - \cos \Delta\phi_{\text{heater}}}{2}, \quad (8)$$

because the interferometer starts from a π phase offset. This is a shifted cosine that goes from 0 (at $\Delta\phi_{\text{heater}} = 0$) to 1 (at $\Delta\phi_{\text{heater}} = \pi$). Beyond π , the output starts decreasing again as the interference moves out of phase. The full period of the cosine corresponds to a 2π phase change from the heater, after which the pattern repeats.

2.4 Extinction Ratio

The extinction ratio characterizes the modulator’s on/off contrast. Using the output power at maximum transmission P_{max} and at minimum transmission P_{min} , the extinction ratio in decibels is

$$\text{ER}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{max}}}{P_{\text{min}}} \right) \quad [3]. \quad (9)$$

A higher ER is desirable for clearer digital 0/1 levels and lower bit error rates. However, extremely high ER can be limited by device imbalances or loss. Many silicon MZIs report ER

~ 20 dB as a typical specification [3]. In this lab, since the operating wavelength was manually set to an interference null, a very low $P_{\min}$ was expected. The measured device demonstrated an ER significantly above 20 dB. The ER is computed from measured data and factors affecting it (such as finite extinction due to imbalanced splitting or background insertion loss) are discussed. It should be noted that achieving ER beyond ~ 30 dB–40 dB may indicate the measured $P_{\min}$ is approaching the noise floor of the photodetector or optical spectrum analyzer.

3 Experimental Procedure

The experiment was conducted on Wednesday, November 12, 2025, using a silicon photonics MZI chip with an integrated micro-heater on one interferometer arm. The setup included an optical source, an optical spectrum analyzer (OSA), and a source-measure unit (SMU) for electrical biasing.

First, a broadband light source (spanning approximately 1520–1580 nm) was coupled into the MZI’s input waveguide. The output spectrum was recorded with an OSA to observe the characteristic interference fringes of the MZI. From this spectral response, a wavelength corresponding to a transmission minimum (an interference trough) was identified. This step was critical: by operating the laser at a trough, the modulator starts in the off state (destructive interference) and will increase in output with applied phase shift. The chosen wavelength was around the 1550 nm telecom band (exact value set to the nearest trough in the fringe pattern). The free spectral range (FSR) of the MZI (spacing between adjacent fringes) was also noted to roughly estimate how much wavelength shift corresponds to a given phase change.

Next, the optical source was switched to a tunable laser set at the selected trough wavelength. The laser provided a continuous-wave input with about 0 dBm launch power into the chip. A Keithley source-measure unit was then used to drive the MZI’s micro-heater. The SMU was configured in voltage source mode, stepping the DC voltage from 0 V upward in increments of approximately 0.5 V, while measuring the resulting current. At each bias point, the output optical power from the MZI was measured. A calibrated photodetector or the OSA in peak-hold mode was used to record the power in dBm. This yielded a set of data points: heater voltage (and current, hence electrical power) versus optical output power.

In addition, the phase shift was verified via spectral peak shifts. At a few intermediate bias points, the OSA was used to capture the shifted interference spectrum. As the heater introduced phase delay, the entire interference fringe pattern shifted in wavelength. By measuring the displacement of a particular spectral peak (or trough) relative to its original position at 0 V, the fraction of a 2π phase change that had occurred was inferred. For example, if a trough moved half the FSR at a certain bias, that indicated a π phase shift. This spectral method provided an independent check of the phase versus voltage relationship. In practice, due to time constraints, most phase data were derived from the transfer function, and the spectral method was observed qualitatively (fringes shifting continuously with bias). However, the linearity of phase with electrical power—a hallmark of thermo-optic modulators—was confirmed by noting that equal voltage increments produced approximately equal fractional fringe shifts.

Data were recorded from 0 V up to about 4.5 V, which was sufficient to drive the modulator past its first maximum transmission and partway into the next interference cycle. The current through the heater at maximum bias was about 23 mA, and no damage or burn-out was observed (the heater was within safe operating range). The SMU’s measured voltage and current were used to compute the electrical power $P = VI$ delivered to the heater at each point.

Key equipment used included: a Newport broadband ASE source (C-band), an ANDO optical spectrum analyzer (0.05 nm resolution), an Agilent tunable laser (1520–1570 nm range), a Thorlabs power meter (InGaAs photodiode), and a Keithley 2400 SMU. Fiber-to-chip coupling was done via polarization-maintaining lensed fibers, and polarization was aligned to the TE mode of the waveguides. The chip was placed on a temperature-stabilized stage (approximately

25 °C base) to minimize drift. Before measurements, the tunable laser wavelength was finely adjusted to precisely find the null (minimum output) at 0 V by tuning ± 0.01 nm and monitoring the power meter.

4 Results

4.1 Phase Shift vs. Heater Power

As the heater power was increased, the induced phase shift in the MZI arm grew roughly linearly. Figure 1 shows the extracted optical phase shift (in radians) as a function of electrical heating power. The phase at each bias point was computed from the interference data—specifically, by normalizing the output power and using the relation

$$\cos(\Delta\phi_{\text{heater}}) = 1 - 2P_{\text{norm}}, \quad (10)$$

valid for the starting condition of destructive interference. For small phase shifts, $\Delta\phi$ is proportional to power, as expected from the linear thermo-optic effect.

Using the linear fit, a π rad (≈ 3.14) phase change corresponds to approximately 50–55 mW of electrical power. The data achieve $\Delta\phi \approx \pi$ at ≈ 46.7 mW, indicating the heater required on the order of tens of milliwatts for a half-wave shift. This is an order of magnitude lower than some longer device designs (which needed hundreds of mW for π [7]), suggesting the heater/waveguide is relatively efficient (likely shorter or thermally isolated). Minor deviations from perfect linearity appeared as $\Delta\phi$ approached π —the last few points (above ~ 30 mW) fell slightly under the linear trend. This could be due to thermal saturation or nonlinear increase in waveguide loss at higher temperatures.

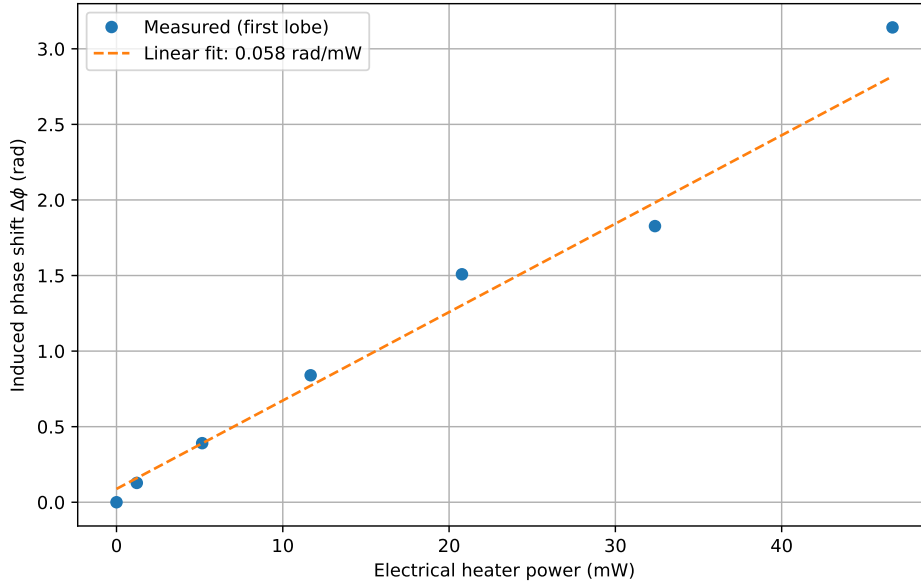


Figure 1: Measured phase shift induced in the MZI versus electrical heating power. The dashed line is a linear fit over the low-to-moderate power region (slope ≈ 0.06 rad/mW), illustrating the approximately linear thermo-optic response. A π phase shift is reached at roughly 47 mW.

4.2 MZI Transfer Function (Output vs. Voltage)

The optical output power (normalized) as a function of heater drive voltage is plotted in Figure 2. The MZI’s transfer function exhibits the expected cosine-shaped response. Starting at 0 V, the

output was near zero (limited by detector noise). As voltage increased, the output rose, reaching a maximum at approximately 3.0 V. This corresponds to the point of π phase difference between the arms (i.e., $V_\pi \approx 3.0$ V for this device). Beyond this voltage, further increases caused the output to decrease, demonstrating the periodic nature of interference.

The measured points are compared to an ideal cosine transfer function (dashed curve) assuming a linear phase–power relation. The agreement is good: the real device reaches about 95–100% of the expected fringe visibility at the peak and closely follows the cosinusoidal trend. Small discrepancies are observed: the measured roll-off beyond 3 V is steeper than ideal, and the symmetry is not perfect—likely because the exact V_π was around 2.9 V (between measurement steps) and there may be slight loss differences in the arms. Nonetheless, the modulator transfer function confirms a half-wave voltage of approximately 3 V and an interference fringe pattern consistent with theory. The device would be biased at $V_\pi/2 \approx 1.5$ V for any small-signal modulation, where the slope dP/dV is maximal and approximately linear.

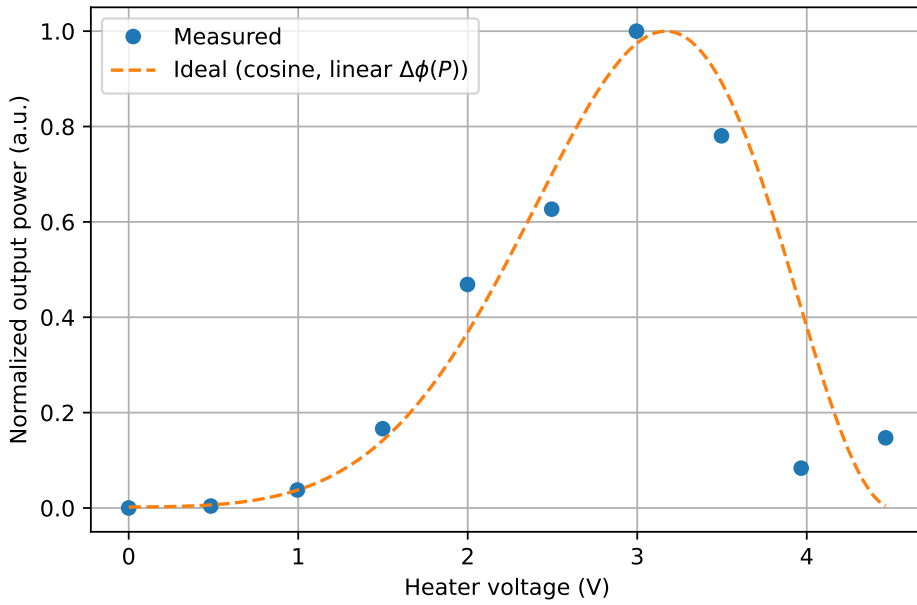


Figure 2: Normalized MZI output power versus heater voltage (transfer function). Points are measured data; the dashed curve is the expected cosine response using a linear $\Delta\phi(P)$ model.

From Figure 2, the extinction ratio of the modulator was determined. The maximum output was about -26.4 dBm, and the minimum output was about -60 dBm. This yields

$$\frac{P_{\max}}{P_{\min}} \approx 2.3 \times 10^3, \quad (11)$$

i.e., an extinction ratio $ER \approx 33.6$ dB. This is a very high contrast; it means the “off” power is more than 2000 times smaller than the “on” power. Such a high ER likely partially reflects the noise floor—in some cases deeper apparent extinction can be recorded when the transmitted minimum power approaches the instrument’s detection limit [4]. Nonetheless, achieving > 30 dB indicates the MZI arms were well balanced in amplitude and phase.

The half-wave voltage was extracted from the first transmission maximum. Interpolating between data points gives $V_\pi \approx 2.9 - 3.0$ V. The corresponding heater power was

$$P_\pi = V_\pi I_\pi \approx (3.0 \text{ V})(15.6 \text{ mA}) \approx 46.8 \text{ mW}, \quad (12)$$

consistent with the phase-versus-power analysis.

Additionally, using the spectral method, it was noted that at $V \approx 3\text{ V}$ the interference fringes on the OSA shifted by roughly one free spectral range (FSR). One FSR shift corresponds to a 2π phase change. Because the measurement started at a trough (π out of phase), a π relative change produced the peak (maximum output) in the monitored port. Thus, both the direct power measurement and spectral observation are consistent with the extracted π phase shift and the half-wave switching voltage.

5 Discussion

The experimental data demonstrate the core principles of thermo-optic MZI modulation and are largely consistent with theoretical expectations.

The linearity of phase shift with electrical power (Figure 1) validates that, to first order, $\Delta\phi \propto P$ for the heater. Since

$$\Delta\phi = \frac{2\pi}{\lambda}(dn/dT) \Delta T L, \quad (13)$$

and, for a given heater, ΔT scales roughly with input power (assuming constant thermal resistance), a linear phase-power relationship is expected. The measured slope ($\sim 0.06\text{ rad/mW}$) implies that approximately 16–17 mW is required for one radian of phase shift. With knowledge of the heater length, an effective (dn/dT) or thermal efficiency could be extracted, but here only empirical numbers are reported. The π shift at $\sim 47\text{ mW}$ is within a reasonable range for on-chip thermo-optic phase shifters. Optimized devices in the literature achieve π with much less power (e.g., 12.7 mW [5]) by suspending the waveguide to thermally isolate it, whereas others require more power depending on design [7]. The device tested here appears moderately thermally isolated.

The transfer function (Figure 2) shows a clear cosine-like behavior. The very high extinction ratio observed is notable. An ideal balanced MZI with 50/50 couplers and identical arm losses can, in theory, achieve infinite extinction at perfect destructive interference. In practice, imbalance and stray reflections limit the extinction. The 33–34 dB ER implies the device was very well balanced; a small residual output at the null likely arises from slight coupler imbalance or arm loss differences. It is also possible that the measured minimum power was near the noise floor, in which case the true ER could be even higher. In modulator terms, the off state is extremely dark, which is beneficial for intensity modulators and switches. However, chasing extremely high ER can make the device more sensitive to thermal drift or wavelength changes. In this experiment, a stable temperature and fixed wavelength were maintained to sustain the deep null [8].

The measured V_π of $\sim 3\text{ V}$ ($\sim 46\text{ mW}$) can be compared to theoretical or datasheet values. Assuming a heater resistance of $\sim 190\ \Omega$ (from $3\text{ V}/15.6\text{ mA}$), this power implies a modest temperature rise. The relatively low V_π is convenient for driver electronics (few-volt drive). In contrast, high-speed depletion-type modulators in silicon often quote $V_\pi L$ products (e.g., 6 V cm), so a centimeter-length device might need 6 V for π [1]. The thermo-optic modulator achieves π in a shorter length at 3 V, highlighting the strength of the thermo-optic effect, albeit with continuous power dissipation.

Minor differences between the measured and ideal transfer curves can arise from:

- Device imbalance (unequal arm loss or non-ideal coupler splitting).
- Thermal crosstalk or resistance change in the heater causing a slightly nonlinear phase versus voltage relation.
- Wavelength drift of the laser or global chip temperature drift during the voltage sweep.
- Instrumentation limits (e.g., OSA resolution and detector noise floor).

Despite these effects, the agreement with the expected $(1 - \cos \phi)/2$ form confirms that a one-arm heating model accurately describes the device.

From a modulation perspective, biasing at quadrature ($V_\pi/2$) is essential for linear operation. Around this point, the slope of $P_{\text{out}}(V)$ is steepest and locally linear. If a small AC signal is applied on top of a 1.5 V DC bias, the output intensity would vary approximately proportionally to that signal. Biasing at 0 V or 3 V, by contrast, would yield low sensitivity or strong nonlinearity. This is analogous to biasing an electronic amplifier in the middle of its transfer curve to minimize distortion [8].

Thermally, the modulator is relatively slow. Thermo-optic phase shifters are limited by heat diffusion; typical switching times for similar devices are on the order of tens to hundreds of microseconds [5, 6], far slower than carrier-depletion or electro-optic modulators operating in the tens-of-gigahertz regime [1]. Thus, thermo-optic MZIs are usually employed for low-frequency control (e.g., filter tuning or on-chip reconfiguration) instead of direct high-speed data modulation. In an applied context, continuously dissipating ~ 47 mW to hold a π shift would generate heat and potential thermal crosstalk in a dense photonic circuit. For lab demonstration and for functions like switching matrices or interferometer trimming, however, thermo-optic MZIs are practical and widely used.

Sources of error include uncertainty in P_{min} (limited by detector noise and stray light), possible wavelength drift, global thermal drift of the chip, SMU measurement uncertainty (on the order of ± 0.1 mA), and finite OSA wavelength resolution. These uncertainties do not significantly alter the main conclusions.

6 Conclusion

Thermal phase modulation in a silicon Mach–Zehnder interferometer was successfully demonstrated and the device’s performance characterized. By heating one arm of the MZI, a variable phase difference was induced and the output intensity was controlled via optical interference. The primary goals—measurement of phase shift versus heater power and recording of the intensity transfer function—were achieved.

A half-wave switching voltage $V_\pi \approx 3$ V was found, corresponding to approximately 47 mW of electrical power for a π phase shift. The MZI exhibited a classic cosine response, with maximum transmission when the arm phase difference reached 2π (full constructive interference). An extinction ratio of approximately 33 dB was achieved, indicating excellent interferometric contrast (off-state optical power on the order of 1 nW versus on-state ~ 2.3 μ W). This exceeds typical values and highlights the high quality of the MZI splitter/combiners and arm balance. Minor deviations from ideal behavior (such as slight asymmetry beyond the first fringe and a small residual output at null) are attributed to practical imperfections and noise limits.

One key takeaway is that thermo-optic MZIs are power-hungry compared to other modulation methods. The tens of milliwatts needed for a single switch would scale to significant power consumption in large arrays of switches. Future work could involve using thermal isolation techniques (e.g., suspending the waveguide or etching trenches) to reduce the power required for a π shift [5]. Another extension would be to measure the dynamic response of the modulator by applying a time-varying current and using a fast photodiode to observe rise/fall times, thereby quantifying the thermal RC time constant [6]. Active cooling or feedback control could also be explored to stabilize the operating point, since thermal devices can drift with ambient conditions. Finally, a comparison between this thermo-optic modulator and a carrier-depletion silicon MZM (covered elsewhere in the course) would be instructive: the latter trades a higher insertion loss and lower ER for gigahertz-speed operation [1].

Overall, the lab provided a comprehensive hands-on understanding of Mach–Zehnder interferometer modulators and the thermo-optic effect, reinforcing theoretical concepts with experimental observation. The measured results confirm that by biasing an MZI at quadrature

and driving it thermally, robust intensity modulation can be achieved in silicon photonics, within the inherent limitations of speed and power associated with thermal tuning.

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